

Lecture Notes 3

Thursday, August 27, 2026 7:12 PM

- 1st project is a video game in Python
 - done in pairs
 - preferable to NOT do in JupiterLab as that's better for data analytics
 - goal is to understand syntax
 - everything covered in first 4-6 weeks must be on this project
 - check-ins done throughout & if done by 4th week, move on to next project, so 4-6 weeks of working on game (i.e. due by 6th week)
 - have general top-down layout/map for game
- Soccer Game
 - Goalkeepers are AI
 - user can't move them
 - 5 v 5
 - user can control other players except GK
 - need collision detection, position
 - user controls player when ball in possession
 - start w/user v computer, then add 1 v 1

UML

Diagram of one class

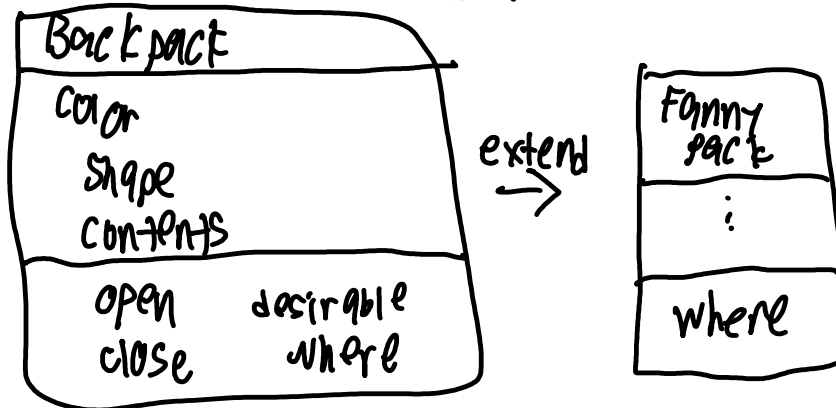


- class name in top of box
- write <<interface>> on

methods

top of interfaces names
— use italics for an
abstract class name

- attributes (optional)
 - should include all fields of object
- operations (methods (optional))
 - may omit trivial (get/set) methods
 - ↳ but don't omit methods from interface!
 - shouldn't include inherited methods



- python doesn't have private var's

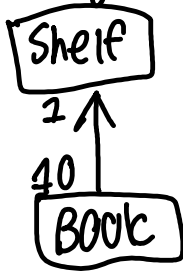
Class Attributes

- attributes (fields, instance variables)
 - visibility:
 - + public
 - # protected
 - private
 - ~ package (default)
 - / derived

Association Types

- aggregation, composition, dependency

⇒ How objects are built is expected for game project



- flowcharts w/dif. symbols

Player

- Marked w/ number 1-10
- Each is represented as a circle
- Highlighter for player user is controlling

Ball

- Represented as black and white
- Smaller than players
- Also a circle

Goals

- 4 goal posts
- 2 cross bars

Field

- Contains all playable objects/lines

Scoreboard

- tracks teams scores

Timer

- tracks time left in game
- accelerated 90 min

Regular Banner

- Contains scoreboard, timer, & team names

Celebration Banner

- Displays when a team scores
- Reset player layout and Ball to middle

Sound

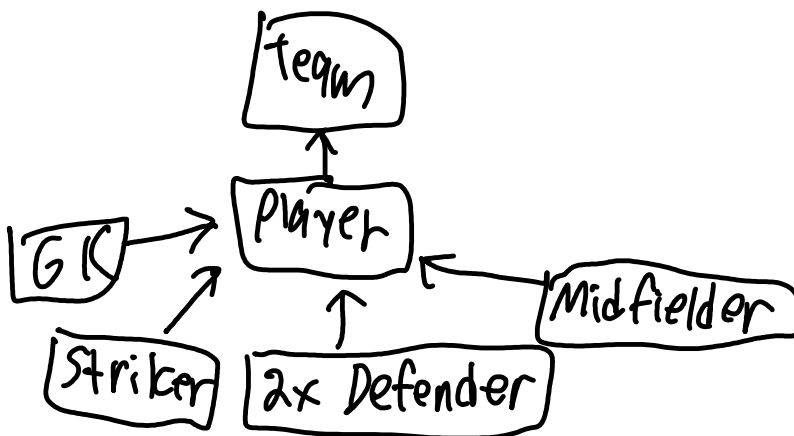
- Each time Ball is kicked
- celebration sound when a team scores

End game Screen

- Displays final score
- stops all objects
- Option to restart match or end game

Team

- contains players



GK

- AI

Striker

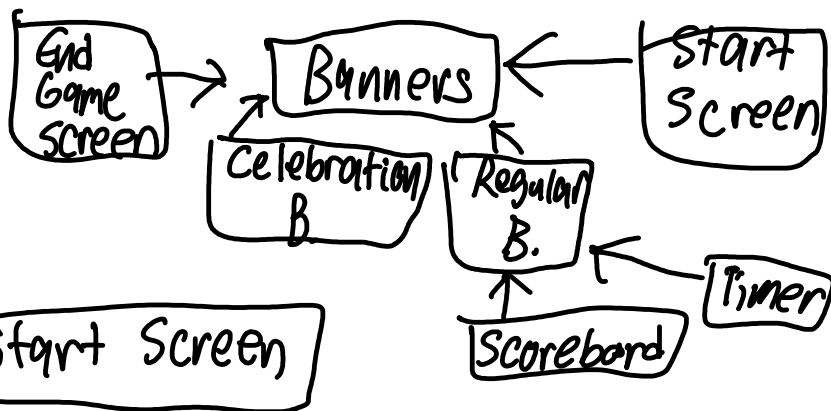
- + shooting

Mid f.

- + passing

2x Def.

- + defense



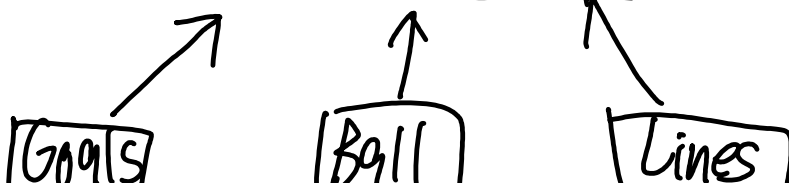
Start Screen

- Start button to begin game
- Option to pick team
- Default Team 1 & Team 2 names
- Give user option to rename both teams
- Exit option

Lines

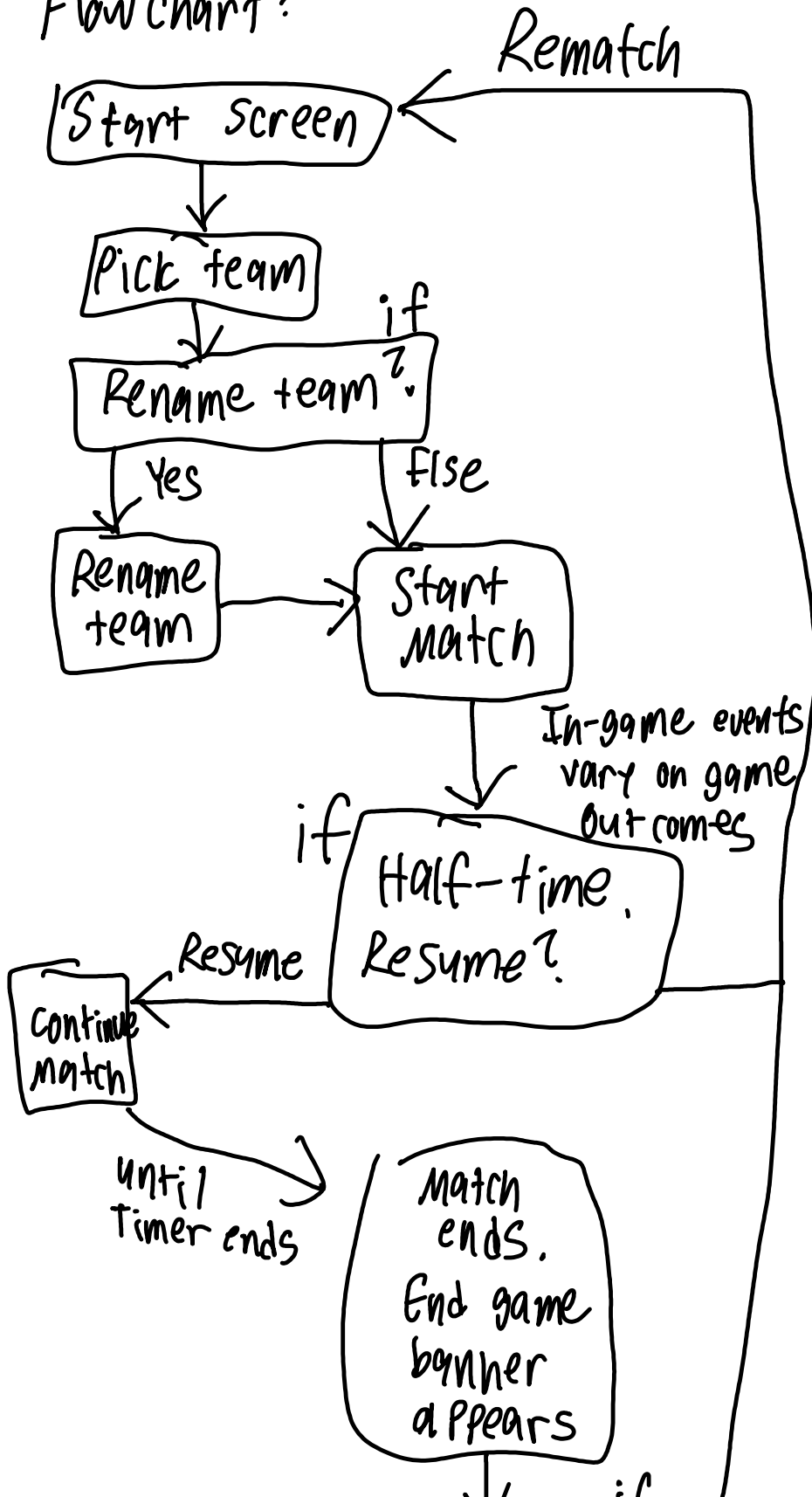
- sets boundaries on field

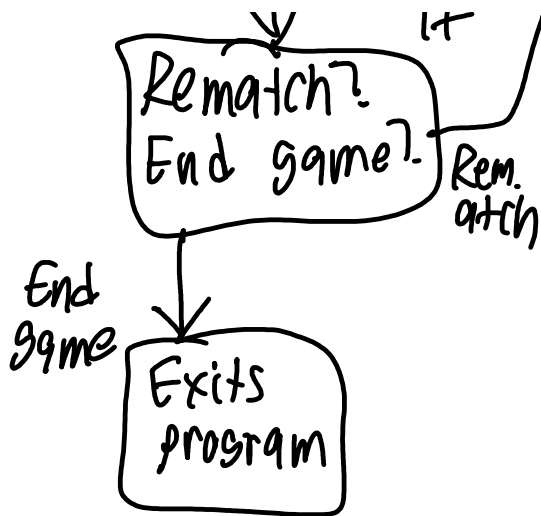
Game Objects



Screen — stand alone object

Flowchart:





class menus
def menu
options (A,
B,
C)

make a menu
function

Choose ...
function

each option calls
a dif. function